

Learned Flow Maps as Hamiltonian Monte Carlo Proposals, and the Symmetry They Require

Abstract

Surrogate Hamiltonian Monte Carlo learns a cheap stand-in for an expensive log-density gradient during warm-up. Every such method we are aware of learns a *vector field* — a gradient, or an energy that is differentiated — and places it inside a symplectic integrator, so advancing L leapfrog steps costs L sequential network evaluations. We propose learning the *flow map* instead, which costs one evaluation regardless of L . Symplectic neural networks are the natural function class, and to our knowledge they have not previously been used as MCMC proposals; the surrogate literature learns fields inside integrators, and the symplectic-network literature predicts dynamics. Making the substitution work requires more than symplecticity. A Hamiltonian flow carries two structures — it preserves the symplectic form, and it is time-reversal symmetric — and a Metropolis correction consumes the second, in the form of an involution. Symplectic networks supply the first by architecture and the second not at all, missing it by $\mathcal{O}(10^{-1})$, so they are not admissible proposals as they stand. Composing the network with its momentum-conjugate inverse restores the missing symmetry exactly, as an algebraic identity in the weights and at no cost in parameters; this is the construction of Valperga et al. (2022), which we transfer from dynamics prediction to sampling and instantiate with closed-form inverses for the LA- and G-SympNet layers. We also find that the conventional all-pairs training set is not merely wasteful but harmful — supervising only the single time offset the sampler queries is $1275\times$ cheaper at $L = 50$ and fits better. On Gaussian-process hyperparameter inference at the configuration of Li et al. (2019), the resulting sampler attains $3.7\times$ the effective samples per second of exact HMC, while every integrator-based surrogate we test is *slower* than exact HMC. We report one cautionary result throughout: an unsymmetrized network posts a competitive 142 ESS/s at a reversibility error of 5×10^{-4} , so effective sample size alone cannot certify a learned proposal.

Keywords: symplectic neural networks, Hamiltonian Monte Carlo, time-reversal symmetry, structure-preserving architectures, surrogate models

1. Learning the flow map instead of the field

Hamiltonian Monte Carlo (Brooks et al., 2011) proposes states by integrating Hamiltonian dynamics and corrects the discretisation error with a Metropolis step. When the target’s gradient is expensive, surrogate methods learn a cheap stand-in during warm-up: a gradient (Li et al., 2019), or a Hamiltonian that is differentiated (Dhulipala et al., 2023). In every such method the learned object is a *vector field* placed inside a symplectic integrator. That arrangement is why they remain asymptotically exact — an inaccurate network lowers the acceptance rate but cannot bias the chain — and it also fixes their cost: advancing L leapfrog steps requires L sequential network evaluations.

We propose learning the flow map $\hat{\phi}(z, \tau)$ directly. Because the elapsed time is an input, a proposal costs one evaluation regardless of L , and the saving grows with trajectory length — the regime adaptive schemes such as NUTS (Hoffman and Gelman, 2011) and

ChEES (Hoffman et al., 2021) push towards. Symplectic neural networks (SympNets; Jin et al., 2020) are the natural function class, sitting in a broader effort to learn Hamiltonian dynamics with architectural constraints (Greydanus et al., 2019; Chen et al., 2019).

To our knowledge this bridge has not been crossed. The surrogate-HMC literature learns fields inside integrators; the symplectic-network literature predicts dynamics and does not treat sampling. The requirement we identify below is therefore a design constraint on a construction we are proposing, not a correction to published work.

Contributions.

- We propose learned symplectic flow maps as HMC surrogates, motivated by an $\mathcal{O}(1)$ rather than $\mathcal{O}(L)$ per-proposal cost.
- We identify what the substitution requires. Replacing the integrator forfeits the correctness it was supplying: a SympNet satisfies volume preservation but violates involutivity by $\mathcal{O}(10^{-1})$, and no amount of training data repairs it (Table 1). The symmetrised map of Valperga et al. (2022) restores it exactly, and we give closed-form inverses for the LA- and G-SympNet layers so the construction is exact rather than approximate.
- We find the conventional all-pairs training set actively harmful: supervising only the offset the sampler queries is $1275\times$ cheaper at $L = 50$ and yields *higher* effective sample size (Appendix B).
- Empirically the symmetrised flow map attains $3.7\times$ the ESS/s of exact HMC on GP hyperparameter inference, where every integrator-based surrogate is slower than exact HMC (Figure 1).

2. What a deterministic proposal requires

Let $\pi(z) \propto e^{-H(z)}$ on $\mathbb{R}^{2D}$ with $H \circ R = H$, where $R(q, p) = (q, -p)$. A Metropolis–Hastings scheme proposing $z' = T(z)$ deterministically and accepting with probability $\min\{1, e^{H(z)-H(Tz)}\}$ leaves π invariant provided

$$(V) \quad |\det \nabla T(z)| = 1, \tag{1}$$

$$(I) \quad (R \circ T) \circ (R \circ T) = \text{Id}, \tag{2}$$

volume preservation and involutivity of $R \circ T$ (Green, 1995; Tierney, 1998). Leapfrog integration of any position-dependent force satisfies both identically — which is exactly what a flow map gives up. That neural proposals must be engineered to satisfy these conditions is established (Song et al., 2017; Levy et al., 2018), and Spanbauer et al. (2020) give a general construction from volume-preserving involutions.

A SympNet composes layers that update either q or p using a function of the other, unmodified half of the state; for the gradient module in “up” mode, $(q, p) \mapsto (q, p + \tau K^\top a \sigma(Kq + b))$. Each layer has unit-determinant Jacobian, so (1) holds for any weights. Condition (2) concerns how the map interacts with R , and no layer constrains it.

Closed-form inverses. The shear structure that makes SympNets symplectic also makes them exactly invertible: in an “up” layer the position passes through untouched, so the increment added to p can be recomputed from the output and subtracted. Composing layer inverses in reverse order gives Φ_t^{-1} exactly; we measure $\|\Phi_t^{-1}\Phi_t z - z\|_\infty \approx 10^{-15}$. Valperga et al. (2022) build their reversible architecture from Hénon maps and coupling layers; the SympNet modules admit the same treatment, which lets the construction apply to networks already in use.

The symmetrised map. Following Valperga et al. (2022),

$$\Psi_t := R \circ \Phi_{t/2}^{-1} \circ R \circ \Phi_{t/2}. \quad (3)$$

Proposition 1 *If $\Phi_{t/2}$ is invertible then Ψ_t satisfies (2); if it is additionally symplectic then Ψ_t satisfies (1). The chain with proposal Ψ_t therefore leaves π invariant for arbitrary weights. Moreover if Φ_s is the exact flow of a momentum-even Hamiltonian then $\Psi_t = \Phi_t$, so the construction does not perturb an exact model.*

Proofs are in Appendix A. Two consequences are worth stating. The map costs two network evaluations and *no additional parameters*, so a Ψ built from an n -block network has the effective depth of a plain $2n$ -block network with half the parameters — comparisons must control for effective depth, not block count. And the property survives composition: if $R\Psi R = \Psi^{-1}$ then $(R\Psi^k)^2 = \text{Id}$ for every k , so multi-step trajectories remain valid proposals.

What the sampler queries. Flow maps are conventionally trained on all pairs (x_i, x_j) , $i < j$, from each warm-up trajectory, giving $\mathcal{O}(L^2)$ examples. But the sampler evaluates $\hat{\phi}$ at exactly one offset, $\tau = L\epsilon$. With finite capacity, fitting times the sampler never queries takes capacity from the one it does; Appendix B shows this costs both compute and accuracy.

3. Results

Models are trained on warm-up trajectories from exact HMC and used as proposals, corrected with the exact Hamiltonian, under a shared optimisation protocol so comparisons reflect architecture rather than tuning. Timings use an isolated GPU.

Table 1 evaluates (1) and (2) at random initialisation, emphasising that these are properties of the construction rather than of training. Figure 1 reports GP hyperparameter inference at the configuration of Li et al. (2019) ($n = 500$, Matérn $\nu = 3/2$, $\epsilon = 0.05$, $L = 20$, 10^4 samples), where each gradient costs an $\mathcal{O}(n^3)$ Cholesky. The symmetrised flow map reaches 177 ESS/s against exact HMC’s 48; the gain is wall-clock rather than sample quality, since HMC attains higher raw ESS (18 634 versus 11 609) in $6\times$ the time. Every integrator-based surrogate is slower than exact HMC (0.41–0.83 $\times$). The advantage grows with L , and the integrator-based surrogate degrades sharply as L grows because it applies its learned field L times and compounds error at each step, whereas the flow map applies once.

Table 1: Validity conditions at random initialisation ($D = 2$, $L = 5$, $\epsilon = 0.1$, double precision). Volume preservation is $\max |\det \nabla T| - 1$; involutivity is $\max \|(RT)^2 z - z\|_\infty$.

proposal map	$ \det \nabla T - 1$	involution error	valid
Exact leapfrog (HMC)	$2 \cdot 10^{-16}$	$3 \cdot 10^{-16}$	yes
Learned field + leapfrog	$8 \cdot 10^{-16}$	$9 \cdot 10^{-16}$	yes
LA-SympNet (plain)	$1 \cdot 10^{-15}$	$2.4 \cdot 10^{-1}$	no
G-SympNet (plain)	$7 \cdot 10^{-16}$	$1.2 \cdot 10^{-1}$	no
LA-SympNet (symmetrised)	$2 \cdot 10^{-15}$	$4 \cdot 10^{-16}$	yes
G-SympNet (symmetrised)	$7 \cdot 10^{-16}$	$1 \cdot 10^{-16}$	yes

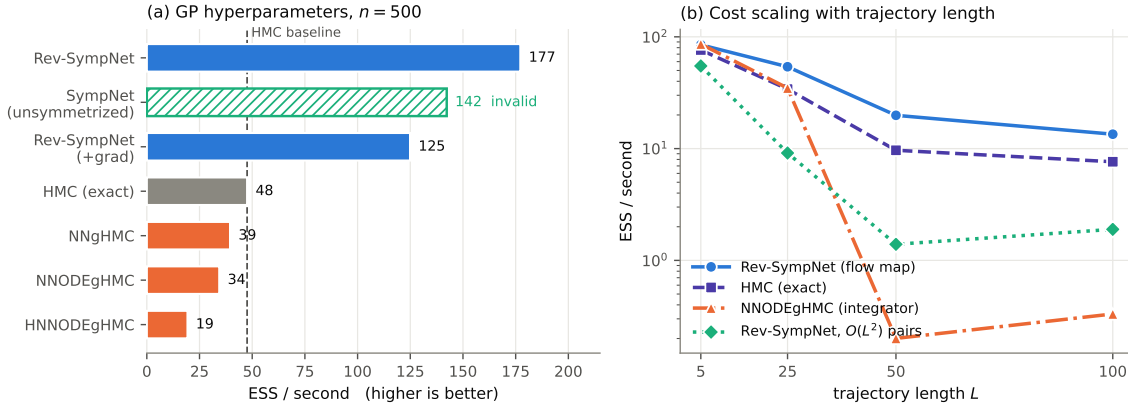


Figure 1: (a) GP hyperparameter inference; best value over training budgets. The symmetrised flow map attains $3.7\times$ the ESS/s of exact HMC, while every integrator-based surrogate is slower than exact HMC. The unsymmetrised network (hatched) posts a competitive number at a reversibility error of 5×10^{-4} and is not a valid sampler. (b) ESS/s against trajectory length: the flow map’s advantage grows with L , the integrator-based surrogate collapses, and training on $O(L^2)$ pairs erases the gain.

Effective sample size does not certify a proposal. In Figure 1 the unsymmetrised network attains 142 ESS/s, which would read as a $3\times$ improvement over exact HMC, at a reversibility error of 5×10^{-4} . A map with a large involution defect can produce weakly autocorrelated paths while targeting the wrong distribution; ESS measures autocorrelation, not correctness. We note this because the artefact is easy to mistake for a result, and we recommend reporting a structural diagnostic, or a two-sample distance against a reference chain, alongside any ESS for a learned proposal.

4. Limitations

Our targets are low-dimensional (two to thirty parameters). A flow map must represent a map on the full $2D$ -dimensional phase space, whereas gradient-learning surrogates fit a field on the D -dimensional position space alone, and we would expect that gap to widen with dimension. Our speedups are measured on GPU, where a 500×500 Cholesky costs about 4 ms and the gradient is not the dominant per-step cost; Li et al. (2019) report $39.3\times$ for a gradient-learning surrogate on their hardware, which we do not reproduce ($0.83\times$) at matched n and L . Our advantage comes from a different mechanism — collapsing L sequential operations into one — and the two numbers are not comparable. Finally, supervising only the queried offset fixes the trajectory length at training time; adaptive-length schemes would require the $\mathcal{O}(L)$ construction, at a measurable cost in fit. Making trajectory length adaptive for a flow map, where length carries no per-proposal cost, is the natural next question.

Appendix A. Proof of Proposition 1

Write $\Phi := \Phi_{t/2}$. Since $R^2 = \text{Id}$, $R \circ \Psi_t = \Phi^{-1} \circ R \circ \Phi$, a conjugation of the involution R , hence an involution: $(R\Psi_t)^2 = \Phi^{-1}R\Phi\Phi^{-1}R\Phi = \Phi^{-1}R^2\Phi = \text{Id}$, which is (2). For (1), R is anti-symplectic, so $R\Phi^{-1}R$ is symplectic and Ψ_t , a composition of two symplectic maps, is symplectic; hence $|\det \nabla \Psi_t| = 1$. Neither argument uses any property of Φ beyond invertibility and symplecticity.

For consistency, reversibility of an exact flow with momentum-even H gives $R \circ \Phi_s \circ R = \Phi_{-s} = \Phi_s^{-1}$, so $\Phi_{t/2}^{-1} = R\Phi_{t/2}R$; substituting into (3) and using $R^2 = \text{Id}$ gives $\Psi_t = \Phi_{t/2} \circ \Phi_{t/2} = \Phi_t$. Equation (3) is the adjoint (symmetrisation) composition of geometric integration (Hairer et al., 2006), with the adjoint realised as $R \circ \Phi^{-1} \circ R$.

Appendix B. Pair construction

Table 2 compares supervising all pairs, all offsets from x_0 , and only (x_0, x_L) , on a Gaussian target with the symmetrised flow map.

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Table 2: Supervising only the queried offset is dramatically cheaper *and* fits better as L grows.

L	pairs	per traj.	total	train (s)	ESS
5	all	15	9 000	10.0	762
5	from x_0	5	3 000	7.8	763
5	endpoint	1	600	7.7	709
25	all	325	195 000	72.0	352
25	from x_0	25	15 000	23.3	348
25	endpoint	1	600	20.1	1175
50	all	1275	765 000	72.4	429
50	from x_0	50	30 000	37.3	267
50	endpoint	1	600	36.3	711

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